

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination – 2022 Course: B. Tech. Branch : Civil Engineering Semester : III Subject Code & Name: BTCVC304 Hydraulics-I Max Marks: 60 Date:15-03-2023 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Explain the different applications in Hydraulics Engineering	CO2	6
B)	A Liquid has a Specific gravity of 1.9 and kinematic Viscosity of 6 stokes. What is its dynamic viscosity?	CO2	6
C)	A rectangular plate 3m long and 1m wide is immersed vertically in water in such a Way that its 3m side is parallel to the water surface and is 1m below it. Find: i) Total Pressure on the plate ii) Position of Centre of Pressure	CO2	6
Q.2	Solve Any Two of the following.		12
A)	What is flow-net? Enumerate the methods of drawing flow nets. Also Define a) Velocity Potential Function b) Stream Function.	CO1	6
B)	List the Flow Measuring Devices and Explain with neat Sketch U-Tube Differential Manometer	CO1	6
C)	The water is flowing through a tapering pipe having diameters 300mm and 150mm at sections 1 and 2 respectively. The discharge through the pipe is 40litre/sec. The section is 10m above datum and section2 is 6m above datum. Find the intensity of Pressure at section2 if that a section1 is 400KN/m ² .	CO2	6

Q. 3	Solve Any Two of the following.		12
A)	What do you meant by Prandtl mixing length theory? Find an expression for shear stress due to Prandtl.	CO4	6
B)	Derive an expression for Euler's equation of motion.	CO4	6
C)	Define Boundary layer and describe effect of pressure gradient on boundary layer separation with neat sketch	CO4	6

Q.4	Solve Any Two of the following.		12
A)	State Buckingham's - theorem. What do you mean by repeating variables? How are the repeating variables selected in dimensional analysis?	CO4	6
B)	Explain the concept of Water Hammer and write a note on Surge Tank	CO3	6
C)	Write short note on Non-dimensional numbers- i) Froude Number ii) Weber Number iii) Reynolds Number	CO4	6

Q. 5	Solve Any Two of the following.		12
A)	Explain With neat sketch Hydraulic gradient and Total energy Line	CO2	6
B)	What is Compound Pipe? What will be loss of head when pipes are connected in series?	CO3	6
C)	The rate of flow of water through a horizontal pipe is $0.25\text{m}^3/\text{s}$. The diameter of the pipe which is 200mm is suddenly enlarged to 400mm. The pressure intensity in the smaller pipe is $11.772 \times 10^4 \text{N/m}^2$. Determine i) loss of head due to sudden enlargement ii) Pressure intensity in larger pipe	CO3	6
	*** End ***		

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